

(VO3) Software Engineering Team

Description & Eligibility

Create a computer software application that performs tasks and operations as outlined in the provided topic. Project submissions will consist of software source code and assets in addition to a functional, executable version of the application. This national event will be submitted and judged virtually. Awards will be presented at the National Leadership Conference. Any middle level, secondary or postsecondary contestant may enter this event. A team will consist of 2-4 contestants. There are no restrictions on the number of entries per chapter or per state. **Teams who do not submit an entry following the topic listed below or do not follow the Copyright and/or Fair Use Guidelines will be disqualified.**

Topic

Create a time-travel puzzle adventure game titled “ChronoQuest: Fractures in Time.” Players are part of a secret organization tasked with repairing the timeline after mysterious anomalies begin altering history. They must travel to different eras—Ancient Egypt, Medieval Europe, the Space Age, and more—to solve puzzles, interact with historical figures, and fix the timeline before reality collapses.

Key features to include:

- Time-travel mechanics with distinct visual and gameplay styles for each era.
- Puzzle-solving that requires logic, pattern recognition, and historical knowledge.
- AI-driven NPCs that react differently based on the timeline’s state.
- Optional co-op mode where players work together across different time periods.
- A branching narrative with multiple endings based on player choices.

Competition Notes

- Contest Registration
 - Advisors register contestants for the event using the Membership Registration System, accessible at <https://register.bpa.org>.
 - Click on the “Conferences” tab and then select the “2026 National Virtual Events”.
 - There will be a \$20 fee for each team entry (*invoiced during event registration*).
 - Event registration can be purchased with your original invoice or with a new invoice after the team registration.
 - The registration deadline is no later than 11:59 p.m. Eastern Time, on December 1, 2025.
 - Contestants participating in the competition must be registered for the event prior to the submission deadline to be eligible.
 - Only the team captain should complete the submission.

Technical Pre-submit Specifications

What to Submit at https://upload.bpa.org	Saved File Name(s)	Deadline
Clickable URL to Project Files, Works Cited and BPA Release Form(s) in one combined PDF file.	VO3-TeamID.pdf	January 15, 2026, 11:59 p.m. Eastern Time

Contest Competencies

- Apply technical skills in the given programming languages to develop the system of applications required.
- Evaluate and delegate responsibilities needed to perform required tasks as a team.
- Develop a project plan and timeline.
- Apply problem solving techniques.
- Implement system analysis and design concepts.
- Use internal programming documentation.
- Use object-oriented programming concepts and techniques.
- Demonstrate knowledge of how data is organized in software development.
- Apply programming concepts such as sequential file access, databases, and I/O operations.
- Use internal and external function and/or procedure calls.

Contest Specifications for this Event

- Interactive game environment.
- Menu interface for non-gameplay user interaction (e.g., starting game, resuming game, choosing levels, exiting the game).
- On-screen HUD for display of vital gameplay information.
- Minimum average framerate of 30 frames per second (FPS).
- Serialization of game state to a database (see below) for saving and resumption of gameplay sessions.
- Values/stats of all game pieces/cards and related data must be stored on a database. Optional databases include Oracle Express 11g, Microsoft SQL Server, Oracle MySQL, PostgreSQL, SQLite, JavaDB, Google Firebase, Amazon Web Services.
- Languages/frameworks permitted include Java, JavaScript, Python, Ruby, Microsoft C#, Objective-C/Cocoa, C++, Apple Swift, Unity, Unreal Engine.
- Additional languages/frameworks/database may be approved by contacting Kelly Williams, kwilliams@bpa.org.
- The use of game development frameworks (e.g., Unity, Unreal Engine, pygame, cocos2d, etc.) is permitted if, and only if, the framework is freely available for educational or nonprofit use. The use of paid proprietary development frameworks is prohibited. Realm Crafter and other “programming light” tools that provide a ready-made gaming engine and/or graphical assets are also prohibited. Any submission using a game development framework must clearly show the extent of original work done via source code comments.
- Use of integrated development environments (e.g., Microsoft Visual Studio, Eclipse, Oracle NetBeans, or Apple XCode) is encouraged.
- Submit the URL to all project documents including, but *not* limited to source code, game manual, project plan, resources, libraries, Works Cited, and BPA Release Form(s) in a combined PDF file to <https://upload.bpa.org> no later than 11:59 p.m. Eastern Time, on January 15, 2026.
- An auto-generated confirmation will be provided upon project submission for the National Leadership Conference; individual confirmation cannot be provided. **No changes can be made to the project after the date of submission.**
- Advisors of qualifying teams will be contacted to reserve a time for a videoconference presentation before a panel of judges. The dates and times for presentation reservations will be announced via email notification. Teams will be assigned to sections prior to the preliminary presentation.
- Length of event: no more than three (3) minutes for setup, no more than ten (10) minutes for the presentation, and no more than five (5) minutes for judges’ questions.
- The top members with the highest cumulative technical and presentation scores will be invited to attend the National Leadership Conference to be recognized for their outstanding efforts and to participate in the National Showcase.

Common Specifications for all BPA Judged Events

- Contestant ID is required for all submissions. Materials from non-registered contestants and/or those missing Contestant ID cannot be accepted.
- The team is responsible for securing a BPA Release Form(s) from any person, including themselves, whose image or work is used in the project.
- The team is responsible for citing all sources, including themselves, on the Works Cited page for any work used on the project. It is the policy of Business Professionals of America to comply with copyright law. The *Style & Reference Manual* contains guidelines for Copyright and/or Fair Use. Contestants will be disqualified for violations of the guidelines.
- Materials must follow the organization’s BPA Graphic Standards and make proper use of the BPA logo and/or organization’s name. Refer to the BPA Graphic Standards in the *Style & Reference Manual*.
- Judges’ comments will be returned digitally through the online judging system at the national level. Materials submitted for technical judging will *not* be returned and will *not* be available at NLC.

(Vo3) Software Engineering Team

Technical Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed the Copyright and/or Fair Use Guidelines.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Submission and Formatting (All points or none are awarded by the technical judge)				
Team submitted one (1) combined PDF file containing the URL to the Project, Works Cited (formatted according to the <i>Style & Reference Manual</i>) and BPA Release Form(s).				/10
Object-Oriented Programming	Below Expectations 0-13 points	Meets Expectations 14-27 points	Above Expectations 27-40 points	Points Awarded
Proper Class Design and Organization	Class structure is disorganized or incomplete. Classes may be missing, overly complex, or lack clear responsibilities. Poor separation of concerns.	Classes are generally well-defined and organized. Responsibilities are mostly clear, with some minor issues in cohesion or structure.	Classes are thoughtfully designed with clear responsibilities, strong cohesion, and logical organization. Demonstrates good architectural planning.	/40
Use of Inheritance	Inheritance is misused or absent. Code may be repetitive or overly complex due to lack of abstraction. Relationships between classes are unclear.	Inheritance is applied appropriately in most cases. Class hierarchies are functional but may lack depth or optimization.	Inheritance is used effectively to promote code reuse and clarity. Class hierarchies are well-structured and demonstrate thoughtful abstraction.	/40
Object-Oriented Programming	Below Expectations 0-10 points	Meets Expectations 11-20 points	Above Expectations 21-30 points	Points Awarded
Code Reuse	Code contains frequent duplication. Reusable components are not implemented, leading to inefficient and hard-to-maintain code.	Code shows some effort to reduce duplication. Basic reuse practices are applied, though opportunities for further abstraction may be missed.	Code is well-structured with high reuse. Functions, classes, and modules are effectively abstracted to minimize duplication and improve clarity.	/30
Use of Encapsulation	Encapsulation is poorly implemented or absent. Data and behavior are exposed unnecessarily, leading to fragile and insecure code.	Basic encapsulation is applied. Most data is protected and access is controlled, though some areas may lack consistency or clarity.	Encapsulation is consistently and effectively used. Data is well-protected, access is thoughtfully managed, and the design promotes maintainability.	/30
Design Analysis	Below Expectations 0-17 points	Meets Expectations 18-34 points	Above Expectations 35-50 points	Points Awarded
Data Flow Diagram(s)	Diagram is missing, unclear, or inaccurate. Data movement and system components are poorly represented or disconnected.	Diagram is present and mostly accurate. Major components and data flows are represented, though some relationships may lack clarity or detail.	Diagram is clear, comprehensive, and well-organized. Accurately represents system components and data flow with logical structure and precision.	/50

(VO3) Software Engineering Team

Technical Scoring Rubric (Con't)

Code Documentation	Below Expectations 0-13 points	Meets Expectations 14-27 points	Above Expectations 27-40 points	Points Awarded
Comment Blocks	Comments are missing or unclear. Code is hard to follow.	Most classes, methods, and logic are explained. Comments are generally helpful but may lack consistency or depth.	Comments are clear, consistent, and enhance understanding. All key components and logic are well-documented.	/40
Code Documentation	Below Expectations 0-10 points	Meets Expectations 11-20 points	Above Expectations 21-30 points	Points Awarded
In-Game Tutorial or Walkthrough	Tutorial is missing, unclear, or difficult to follow. Users may struggle to understand gameplay or features.	Tutorial is present and functional. It covers basic gameplay and features, though it may lack polish or depth.	Tutorial is clear, engaging, and comprehensive. It effectively guides users through gameplay and features with ease and clarity.	/30
Crash Reporting	Below Expectations 0-10 points	Meets Expectations 11-20 points	Above Expectations 21-30 points	Points Awarded
Generation of Crash Reports	Crash reporting is missing or unreliable. Failures are not logged or communicated to the user.	Crash reports are generated and provide basic information. Functionality is present but may lack detail or consistency.	Crash reports are consistently generated with clear, detailed information. Help users and developers understand and address failures effectively.	/30
Crash Reporting	Below Expectations 0-7 points	Meets Expectations 8-13 points	Above Expectations 14-20 points	Points Awarded
Email Crash Report	No option to email crash reports. Users have no way to share failure details with developers.	Email option is available and functional. Basic crash information can be sent, though the process may lack automation or clarity.	Email option is well-integrated and user-friendly. Crash reports are detailed and sent efficiently, supporting quick issue resolution.	/20
Data Driven Design	Below Expectations 0-10 points	Meets Expectations 11-20 points	Above Expectations 21-30 points	Points Awarded
Data Driven Design Runtime Settings	Runtime settings are hardcoded or manually configured. No database integration is evident.	Basic use of a database to manage runtime settings. Functionality is present but may be limited or inconsistently applied.	Runtime settings are fully managed through a database. Implementation is flexible, scalable, and supports dynamic configuration.	/30
Session Data Storage	Session data is not saved or is stored unreliably. Users cannot resume progress or access previous records.	Session data is saved and retrievable. Basic functionality is present, though it may lack robustness or flexibility.	Session data is consistently and securely stored. Users can reliably resume progress, and data is well-integrated into the application workflow.	/30

(VO3) Software Engineering Team

Technical Scoring Rubric (Con't)

Data-driven Game Content	Runtime settings are hardcoded or manually configured. No database integration is used for dynamic control.	Runtime settings are stored in a database and can be modified. Functionality is present but may be limited or inconsistently applied.	Runtime settings are fully integrated with a database. Changes are dynamic, flexible, and enhance user or system control during execution.	/30
Error Handling	Below Expectations 0-7 points	Meets Expectations 8-13 points	Above Expectations 14-20 points	Points Awarded
Proper Use of Error Handling	Error handling is missing or ineffective, resulting in crashes or unpredictable behavior.	Basic error handling covers common issues, though some edge cases may be missed.	Error handling is reliable and consistently improves stability and user experience.	/20
Proper Use of Exception Handling	Exceptions are not handled or are handled incorrectly. Code may crash or behave unpredictably when errors occur.	Exceptions are handled in most cases. Basic try/catch structures are used, though some scenarios may be overlooked or handled ineffectively.	Exception handling is robust and well-structured. Errors are caught gracefully, with meaningful responses that maintain application stability.	/20
Error Handling	Below Expectations 0-8 points	Meets Expectations 9-16 points	Above Expectations 16-25 points	Points Awarded
User Alerts on Error Conditions	Alerts are missing or unclear. Users are not informed about errors or how to respond, leading to confusion or frustration.	Alerts are present and generally informative. Users are notified of errors, though messages may lack clarity or consistency.	Alerts are clear, well-structured, and distinguish between recoverable and non-recoverable errors. They guide users effectively through issues.	/25
Logging	Below Expectations 0-10 points	Meets Expectations 11-20 points	Above Expectations 21-30 points	Points Awarded
Logging System Events	Logging is missing or inconsistent. Events are not recorded or are difficult to trace.	System events are logged to a text file. Logs are generally useful but may lack structure or completeness.	Logging is thorough and well-organized. Events are clearly recorded, aiding effective debugging and system analysis.	/30
Log System Errors to Dedicated Text File	Errors are not logged or logs are incomplete. Troubleshooting is difficult due to lack of recorded information.	Errors are logged to a text file. Logs contain useful information but may lack consistency, structure, or depth.	Error logging is detailed, consistent, and well-structured. Logs provide clear insights that support effective debugging and issue resolution.	/30
Team submission includes a manual containing directions for compiling/building and explains the features of the submission.				/100
Team included the installer for the project application.				/25
TOTAL TECHNICAL POINTS				/660

(Vo3) Software Engineering Team

Gameplay Scoring Rubric

Evaluation Criteria	Below Expectations 0-10 points	Meets Expectations 11-20 points	Above Expectations 21-30 points	Points Awarded
Interface Design	Interface is cluttered, confusing, or inconsistent. Navigation and usability are poor.	Interface is functional and mostly intuitive. Basic design principles are applied, though some areas may lack polish or consistency.	Interface is clean, user-friendly, and visually appealing. Design enhances usability and provides a smooth user experience.	/30
Inclusion of Audio	Audio is missing, poorly implemented, or detracts from the experience.	Audio is present and functional. It supports gameplay but may lack variety, quality, or integration.	Audio is well-integrated, enhances immersion, and complements gameplay. Includes thoughtful use of sound effects and/or music.	/30
Logical Controls	Controls are confusing, unresponsive, or poorly mapped. Gameplay is hindered by lack of intuitive interaction.	Controls are functional and mostly intuitive. Some minor issues may affect user experience or responsiveness.	Controls are smooth, responsive, and well-designed. They enhance gameplay by providing a seamless and intuitive user experience.	/30
Amount of Content	Game content is minimal or repetitive. Lacks variety or depth to keep users engaged.	Game includes a reasonable amount of content. Offers enough variety to support basic gameplay and user interest.	Game is rich in content with diverse features, levels, or challenges. Provides depth and replayability that enhances user engagement.	/30
Lasting Appeal	Game lacks replay value or long-term engagement. Players are unlikely to return after initial use.	Game offers moderate replayability. Content and design encourage some continued interest, though it may fade over time.	Game is highly engaging and replayable. Content, design, and challenge levels encourage repeated play and long-term enjoyment.	/30
TOTAL GAMEPLAY POINTS				/150

(Vo3) Software Engineering Team

Presentation Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Items To Evaluate	Below Expectations 0-5 points	Meets Expectations 6-10	Above Expectations 11-15	Points Awarded
Full Design & Development Process	Process is unclear or incomplete; lacks structure or detail.	Process is explained with basic clarity and covers major phases.	Process is clearly explained with insight into decisions.	/15
Manage Game Data Flow	Data flow is vague or incorrect; lacks technical clarity.	Data flow is described with general accuracy and relevance.	Data flow is clearly explained with technical relevance to gameplay.	/15
Game Design & Mechanics	Game design and mechanics are poorly explained or disconnected.	Game design and mechanics are described with basic clarity and relevance.	Game design and mechanics are clearly explained and well-integrated.	/15
Media Design & Integration	Media elements are mentioned but not explained or justified.	Media elements are described with some connection to gameplay and design.	Media elements are clearly explained and well-integrated into the game's design and experience.	/15
Applied SE Principles	Principles are not mentioned or misapplied.	Basic principles are referenced and generally applied.	Principles are clearly shown and effectively applied in the project.	/15
Game Innovation	Innovation is unclear or unsupported.	Innovation is described with some examples or features.	Innovation is clearly demonstrated with unique features or approaches.	/15
Clear Interview Responses	Responses are vague, incomplete, or off topic.	Responses are mostly clear and relevant.	Responses are confident and show strong understanding.	/15
Team Contribution to Development	Team roles are unclear or unevenly represented.	Each team member's role is described with basic clarity.	Responses are confident and show strong understanding.	/15
Presentation Quality & Style	Presentation lacks organization, clarity, or engagement.	Presentation is organized and communicates key points effectively.	Presentation is polished, engaging, and professionally delivered.	/15
Team Members	Set up lasted no longer than three (3) minutes and the presentation lasted no more than ten (10) minutes.			/5
Timing	At least two original members of the team in attendance at the time of the presentation.			/10
TOTAL PRESENTATION POINTS				/150
Scoring Category				Points Awarded
Technical Scoring Points				/660
Gameplay Points				/150
Presentation Points				/150
TOTAL				/960

Props and/or additional items shall not be used as a basis for scoring.

PRESENTATION WILL BE STOPPED AT TEN (10) MINUTES